

David LeBauer, Ph.D.

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Employment

The LeBauer Approach, LLC

Owner and Principal Scientist

Mar 2024–present

Modeling Lead and Project Manager, California Cropland Measurement and Modeling Framework (Sept 2024–present).

Scientific Advisor, Varaha Climate AG (Jun 2024–present).

Scientific consultant.

Indigo Ag

Staff Scientist and Technical Lead, Model Validation and Calibration Feb 2023–Feb 2024

Technical lead for team that calibrated, validated, and wrote report following Climate Action Reserve Soil Enrichment Protocol to support generation of verified carbon credits.

Wrote science requirements for automated model calibration and validation.

Harmonized calibration and validation datasets of observed soil carbon, greenhouse gas fluxes, and yields; migrated dataset to a relational geospatial database.

University of Arizona, Division of Agriculture, Life and Veterinary Sciences, and Cooperative Extension

Adjunct Data Scientist

Feb 2023–present

Director of Data Science

Jul 2018–Jan 2023

Founded and built a data science support team comprising 4–6 data scientists and software engineers to provide research support for the Division of Agriculture, Life Science, and Cooperative Extension.

Facilitated cross-disciplinary, collaborative research projects through data curation, statistical guidance, grant writing, workshops, and translation of research priorities across domains.

Developed and launched a data science incubator program with outcomes that include preliminary results, upgraded scientific software, published papers, data, and code, and

data visualization dashboards and analysis pipelines.

Developed long-term vision and five-year plan using core USDA Hatch funding alongside over \$2m in extramural funding to support modeling, computing, and informatics projects from ARPA-E, DARPA, USDA, and NSF.

Agribile, Inc. (part-time)

Senior Scientific Advisor

Jun 2015–Jul 2018

Designed and implemented analyses of yield data used to inform seed choice.

Consulted with software engineers and scientists to develop, analyze, and apply a crop model to inform and optimize farms and supply chains.

Taught quantification and propagation of uncertainty in crop models, including soil physical properties, and advised on implementation of these methods.

University of Illinois at Urbana-Champaign

Research Scientist, Carl R. Woese Institute for Genomic Biology

Aug 2012–Jun 2018

Data and Computing Pipeline Lead, TERRA-REF (2015–2019).

Awarded \$1.7m from DOE plus over \$1m in computing and storage from NSF-funded resources to lead a team of three full-time and ten part-time programmers for a sensor-based plant observation data stream.

Coordinated development of software and data products across six collaborating research groups across genomics, physiology, breeding, remote sensing, computer vision, and robotics.

Integrated feedback from end users, an advisory committee, and industry partners into data products and metadata standards.

Published a public domain agricultural data set exceeding 1 PB of sensor, trait, and genomics data.

Scientific Manager, EBI Ecosystem Modeling Program

Aug 2012–Jul 2015

Managed research program predicting yield potential, yield stability, and environmental impacts of bioenergy feedstock production.

Wrote proposals and administered a \$600k annual budget supporting four faculty, three postdocs, one research programmer, a project manager, contractors, and student employees.

Led development and application of simulation, statistical, and informatics software.

Postdoctoral Researcher, Energy Biosciences Institute

Aug 2009–Jul 2012

Created, implemented (as PEcAn), and distributed a software platform and database

to support synthesis of ecophysiological and yield data and model–data synthesis.

Developed protocol and database, and designed a UI to support efficient extraction of data from literature by teams of scientists and technicians.

North Carolina Agricultural and Technical State University

Lab Manager, Mushroom Biology and Fungal Biotechnology Laboratory Jun 2003–Jul 2004

Duke University

Laboratory and Field Technician, Biogeochemistry and Community Ecology Labs Jun 1996–Jul 2001

Education

- 2008** **PhD Earth System Science**, University of California at Irvine
Thesis: ‘Carbon Balance of Ecosystems Under Nitrogen Enrichment’
- 2003** **MS Ecology, Agriculture concentration**, University of California at Davis
- 1998** **BS Biology**, Duke University

Publications

Monographs

LeBauer, David S., Dan Wang, Katherine T. Richter, Carl C. Davidson, and Michael C. Dietze (May 2013). “Facilitating feedbacks between field measurements and ecosystem models”. In: *Ecological Monographs* 83.2, pp. 133–154. ISSN: 00129615. DOI: 10.1890/12-0137.1. URL: <http://www.esajournals.org/doi/abs/10.1890/12-0137.1>.

Journal articles

- Lachowicz, Jennifer, Max J. Feldman, Filipe Inacio Matias, David LeBauer, and Alexander Gregory (2024). “Adoption of unoccupied aerial systems in agricultural research”. In: *The Plant Phenome Journal* 7.1, e20098. DOI: <https://doi.org/10.1002/ppj2.20098>. URL: <https://access.onlinelibrary.wiley.com/doi/abs/10.1002/ppj2.20098>.
- Wheeler, Kathryn I, Michael C Dietze, David LeBauer, Jody A Peters, Andrew D Richardson, Arun A Ross, R Quinn Thomas, Kai Zhu, Uttam Bhat, Stephan Munch, et al. (2024). “Predicting spring phenology in deciduous broadleaf forests: NEON phenol-

- ogy forecasting community challenge”. In: *Agricultural and Forest Meteorology* 345, p. 109810. DOI: 10.1016/j.agrformet.2023.109810.
- Meredith, Laura K, S Marshall Ledford, Kristina Riemer, Parker Geffre, Kelsey Graves, Linnea K Honeker, David LeBauer, Malak M Tfaily, and Jordan Krechmer (2023). “Automating methods for estimating metabolite volatility”. In: *Frontiers in Microbiology* 14. DOI: 10.3389/fmicb.2023.1267234.
- Roy, Bishal, Vasit Sagan, Alifu Haireti, Maria Newcomb, Roberto Tuberosa, David LeBauer, and Nadia Shakoor (2023). “Early Detection of Drought Stress in Durum Wheat Using Hyperspectral Imaging and Photosystem Sensing”. In: *Remote Sensing* 16.1, p. 155. DOI: 10.3390/rs14010155.
- LeBauer, David, Alexander Bucksch, Jennifer Clarke, Jesse Potts, and Sonali Roy (2023). “Increasing racial diversity in the North American Plant Phenotyping Network through conference participation support”. In: *The Plant Phenome Journal* 6.1, e20075. DOI: 10.1002/ppj2.20075.
- Ely, Kim S., Dedi Yang, Shawn P. Serbin, Christian G. Andresen, Yose Angel, Adam Collins, Joan Damerow, Brett Feddersen, Sean Freeman, Peng Fu, Brian Greene, Wouter Hanston, Mark J. Lara, David LeBauer, Trina Merrick, Emily Robles, Alexey Shiklomanov, Matthias Siewert, Charaluka Varadharajan, and Jin Wu (Jan. 2023). *ESS-DIVE Unoccupied Aerial Systems (UAS) Reporting Format v1*. DOI: 10.15485/2204420. URL: <https://www.osti.gov/biblio/2204420>.
- Todd-Brown, Katherine, Rose Z Abramoff, Jeffrey Beem-Miller, Hava K Blair, Stevan Earl, Kristen J Frederick, Daniel R Fuka, Mario Guevara Santamaria, Jennifer W Harden, Katherine A Heckman, Lillian Joan Heran, James R Holmquist, Alison M Hoyt, David H Klinges, David S LeBauer, Avni Malhotra, Shelby C McClelland, Lucas E Nave, Katherine S Rocci, Sean Michael Schaeffer, Shane Stoner, Sophie F von Fromm Natasja van Gestel, and Marisa Loren Younger (2022). “Review and Synthesis: The promise of big soil data, moving current practices towards future potential”. In: *Biogeosciences Discussions*. DOI: 10.5194/bg-2021-323. URL: <https://bg.copernicus.org/preprints/bg-2021-323/>.
- Matias, Filipe Inácio, Andrew Green, Jennifer A Lachowicz, David LeBauer, and Max Feldman (2022). “Bison-Fly: An open-source UAV pipeline for plant breeding data collection”. In: *The Plant Phenome Journal* 5.1, e20048. DOI: 10.1002/ppj2.20048.
- Mpanga, Isaac Kwadwo, Russell Tronstad, Jessica Guo, David Shaner LeBauer, and Omololu John Idowu (2021). “On-farm land management strategies and production challenges in United States organic agricultural systems”. In: *Current Research in Environmental Sustainability* 3, p. 100097. ISSN: 2666-0490. DOI:

<https://doi.org/10.1016/j.crsust.2021.100097>. URL: <https://www.sciencedirect.com/science/article/pii/S2666049021000736>.

- Sagan, Vasit, Maitiniyazi Maimaitijiang, Sidike Paheding, Sourav Bhadra, Nichole Gosselin, Max Burnette, Jeffrey Demieville, Sean Hartling, David LeBauer, Maria Newcomb, et al. (2021). “Data-Driven Artificial Intelligence for Calibration of Hyperspectral Big Data”. In: *IEEE Transactions on Geoscience and Remote Sensing*.
- Davis, Sarah C., John T. Abatzoglou, and David S. LeBauer (2021). “Expanded Potential Growing Region and Yield Increase for Agave americana with Future Climate”. In: *Agronomy* 11.11. ISSN: 2073-4395. DOI: 10.3390/agronomy11112109. URL: <https://www.mdpi.com/2073-4395/11/11/2109>.
- Sahneh, Faryad, Meghan A. Balk, Marina Kisley, Chi-kwan Chan, Mercury Fox, Brian Nord, Eric Lyons, Tyson Swetnam, Daniela Huppenkothen, Will Sutherland, Ramona L. Walls, Daven P. Quinn, Tonantzín Tarín, David LeBauer, David Ribes, Dunbar P. Birnie III, Carol Lushbough, Eric Carr, Grey Nearing, Jeremy Fischer, Kevin Tyle, Luis Carrasco, Meagan Lang, Peter W. Rose, Richard R. Rushforth, Samapriya Roy, Thomas Matheson, Tina Lee, C. Titus Brown, Tracy K. Teal, Monica Papeş, Stephen Kobourov, and Nirav Merchant (May 2021). “Ten simple rules to cultivate transdisciplinary collaboration in data science”. In: *PLOS Computational Biology* 17.5, pp. 1–12. DOI: 10.1371/journal.pcbi.1008879. URL: <https://doi.org/10.1371/journal.pcbi.1008879>.
- Ely, Kim S., Alistair Rogers, Deborah A. Agarwal, Elizabeth A. Ainsworth, Loren Albert, Ashehad Ali, Jeremiah Anderson, Michael J. Aspinwall, Chandra Bellasio, Carl Bernacchi, Steve Bonnage, Thomas N. Buckley, James Bunce, Angela C. Burnett, Florian A. Busch, Amanda Cavanagh, Lucas A. Cernusak, Robert Crystal-Ornelas, Joan Damerow, Kenneth J. Davidson, Martin G. De Kauwe, Michael C. Dietze, Tomas F. Domingues, Mirindi Eric Dusenge, David S. Ellsworth, John R. Evans, Paul P.G. Gauthier, Bruno O. Gimenez, Elizabeth P. Gordon, Christopher M. Gough, Aud H. Halbritter, David T. Hanson, Mary Heskell, J. Aaron Hogan, Jason R. Hupp, Kolby Jardine, Jens Kattge, Trevor Keenan, Johannes Kromdijk, Dushan P. Kumarathunge, Julien Lamour, Andrew D.B. Leakey, David S. LeBauer, Qianyu Li, Marjorie R. Lundgren, Nate McDowell, Katherine Meacham-Hensold, Belinda E. Medlyn, David J.P. Moore, Robinson Negrón-Juárez, Ülo Niinemets, Colin P. Osborne, Alexandria L. Pivovarov, Hendrik Poorter, Sasha C. Reed, Youngryel Ryu, Alvaro Sanz-Saez, Stephanie C. Schmiege, Shawn P. Serbin, Thomas D. Sharkey, Martijn Slot, Nicholas G. Smith, Balasaheb V. Sonawane, Paul F. South, Daisy C. Souza, Joseph Ronald Stinziano, Ellen Stuart-Haëntjens, Samuel H. Taylor, Mauricio D. Tejera, Johan Uddling, Vigdis Vandvik, Charuleka Varadharajan, Anthony P. Walker, Berkley J. Walker, Jeffrey M. Warren, Danielle A. Way, Brett T. Wolfe, Jin Wu, Stan D. Wullschlegel, Chonggang Xu, Zhengbing Yan, and Dedi Yang (2021). “A reporting format for leaf-level gas ex-

change data and metadata”. In: *Ecological Informatics*, p. 101232. ISSN: 1574-9541. DOI: 10.1016/j.ecoinf.2021.101232. URL: <http://www.sciencedirect.com/science/article/pii/S1574954121000236>.

- Field, John L., Tom L. Richard, Erica A. H. Smithwick, Hao Cai, Mark S. Laser, David S. LeBauer, Stephen P. Long, Keith Paustian, Zhangcai Qin, John J. Sheehan, Pete Smith, Michael Q. Wang, and Lee R. Lynd (2020). “Robust paths to net greenhouse gas mitigation and negative emissions via advanced biofuels”. In: *Proceedings of the National Academy of Sciences* 117.36, pp. 21968–21977. ISSN: 0027-8424. DOI: 10.1073/pnas.1920877117. eprint: <https://www.pnas.org/content/117/36/21968.full.pdf>. URL: <https://www.pnas.org/content/117/36/21968>.
- Maimaitijiang, M., V. Sagan, H. Erkbol, J. Adrian, M. Newcomb, D.S. LeBauer, D. Pauli, N. Shakoor, and T. C. Mockler (2020). “UAV-Based Sorghum Growth Monitoring: A Comparative Analysis of LIDAR and Photogrammetry”. In: *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences* V-3-2020, pp. 489–496. DOI: 10.5194/isprs-annals-V-3-2020-489-2020. URL: <https://www.isprs-ann-photogramm-remote-sens-spatial-inf-sci.net/V-3-2020/489/2020/>.
- Fer, Istem, Anthony K. Gardella, Alexey N. Shiklomanov, Eleanor E. Campbell, Elizabeth M. Cowdery, Martin G. De Kauwe, Ankur Desai, Matthew J. Duveneck, Joshua B. Fisher, Katherine D. Haynes, Forrest M. Hoffman, Miriam R. Johnston, Rob Kooper, David S. LeBauer, Joshua Mantooth, William J. Parton, Benjamin Poulter, Tristan Quaife, Ann Raiho, Kevin Schaefer, Shawn P. Serbin, James Simkins, Kevin R. Wilcox, Toni Viskari, and Michael C. Dietze (2020). “Beyond ecosystem modeling: A roadmap to community cyberinfrastructure for ecological data-model integration”. In: *Global Change Biology* 27.1, pp. 13–26. DOI: <https://doi.org/10.1111/gcb.15409>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/gcb.15409>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/gcb.15409>.
- Selby, Peter, Rafael Abbeloos, Jan Erik Backlund, Martin Basterrechea Salido, Guillaume Bauchet, Omar Benites-Alfaro, Clay Birkett, Viana C Calaminos, Pierre Carceller, Guillaume Cornut, Bruno Vasques Costa, Jeremy D Edwards, Richard Finkers, Star Yanxin Gao, Mehmood Ghaffar, Philip Glaser, Valentin Guignon, Puthick Hok, Andrzej Kilian, Patrick König, Jack Elendil B Lagare, Matthias Lange, Marie-Angélique Laporte, Pierre Larmande, David S. LeBauer, David Lyon, David Marshall, Dave Matthews, Iain Milne, Naymesh Mistry, Nicolas Morales, Lukas Mueller, Pascal Neveu, Evangelia Papoutsoglou, Brian Pearce, Ivan Perez-Masias, Cyril Pommier, Ricardo H Ramirez-Gonzalez, Abhishek Rathore, Angel M Raque, Sebastian Raubach, Trevor Rife, Kelly Robbins, Mathieu Rouard, Chaitanya Sarma, Uwe Scholz, Guilhem Sempéré, Paul Shaw, Reinhard Simon, Nahuel Soldevilla, Gordon Stephen, Qi Sun, Clarysabel Tovar, Grzegorz Uszynski, and Maikel Verouden (2019). “BrAPI - an Application Programming Inter-

- face for Plant Breeding Applications”. In: *Bioinformatics*. ISSN: 1367-4803. DOI: 10.1093/bioinformatics/btz190. URL: <https://doi.org/10.1093/bioinformatics/btz190>.
- Babaeian, Ebrahim, Paheding Sidike, Maria S. Newcomb, Maitiniyazi Maimaitijiang, Scott A. White, Jeffrey Demieville, Richard W. Ward, Morteza Sadeghi, David S. LeBauer, Scott B. Jones, Vasit Sagan, and Markus Tuller (2019). “A New Optical Remote Sensing Technique for High-Resolution Mapping of Soil Moisture”. In: *Frontiers in Big Data* 2, p. 37. ISSN: 2624-909X. DOI: 10.3389/fdata.2019.00037. URL: <https://www.frontiersin.org/article/10.3389/fdata.2019.00037>.
- LeBauer, David S, Rob Kooper, Steven P Long, Patrick J Mulrooney, Scott Rohde, Dan Wang, and Michael C Dietze (2018). “BETYdb: A Yield, Trait and Ecosystem Service Database Applied to Second Generation Bioenergy Feedstocks”. In: *Global Change Biology Bioenergy*. DOI: 10.1111/gcbb.12420.
- Jaiswal, Deepak, Soren Larsen, Amanda De Souza, David S LeBauer, Fernando E Miguez, Gerd Sparovek, Marcos S Buckeridge, German Bollero, and Steven P Long (2017). “Brazilian Sugarcane Ethanol As an Expandable Green Alternative to Crude Oil Use”. In: *Nature Climate Change*. DOI: 10.1038/NCLIMATE3410.
- Black, Christopher K, Michael D. Masters, David S. LeBauer, Kristina J. Anderson-Teixeira, and Evan H. DeLucia (2017). “Root volume distribution of maturing perennial grasses revealed by correcting for minirhizotron surface effects”. In: *Plant and Soil*. DOI: 10.1007/s11104-017-3333-7.
- Hart, Edmund, Pauline Barmby, David S LeBauer, Francois Michonneau, Sarah Mount, Patrick Mulrooney, Timothee Poisot, Kara H Woo, Naupaka Zimmerman, and Jeffrey W Hollister (2016). “Ten simple rules for digital data storage”. In: *PLOS Computational Biology*. DOI: 10.1371/journal.pcbi.1005097.
- Anderson-Teixeira, Kristina J, Maria Wang, Jennifer C McGarvey, and David S LeBauer (2016). “Carbon dynamics of mature and regrowth tropical forests derived from a pantropical database (TropForC)”. In: *Global Change Biology*. DOI: 10.1111/gcb.13226.
- Zhu, Xin-Guang, Jonathan P. Lynch, David S. LeBauer, Andrew J. Millar, Mark Stitt, and Stephen P. Long (2015). “Plants in silico: Why, Why Now and Framework - An integrative platform for plant systems biology research.” In: *Plant, Cell & Environment*. DOI: 10.1111/pce.12673. URL: <http://dx.doi.org/10.1111/pce.12673>.
- Wang, Dandan; Deepak; Jaiswal, David; LeBauer, Timothy; Wertin, German; Bollero, Andrew; Leakey, and Steve Long (2015). “A physiological and biophysical model of coppice willow (*Salix* spp.) production for the contiguous USA in current and future climate scenarios”. In: *Plant, Cell & Environment*. URL: <http://dx.doi.org/10.1111/pce.12556>.

- Davis, Sarah, Ray Ming, David S LeBauer, and Stephen Long (2015). “Toward systems-level analysis of agricultural production from Crassulacean Acid Metabolism (CAM): scaling from cell to commercial production”. In: *New Phytologist*. DOI: 10.111/nph.13522. URL: <http://dx.doi.org/10.111/nph.13522>.
- Davis, Sarah C, David S LeBauer, and Stephen P Long (2014). “Light to liquid fuel: theoretical and realized energy conversion efficiency of plants using Crassulacean Acid Metabolism (CAM) in arid conditions.” In: *Journal of experimental botany* 65.13, pp. 3471–3478. ISSN: 1460-2431. DOI: 10.1093/jxb/eru163. URL: <http://www.ncbi.nlm.nih.gov/pubmed/24744431>.
- Wang, Dan, David S LeBauer, and Michael Dietze (June 2013). “Predicting yields of short-rotation hybrid poplar (Populus spp.) for the United States through model–data synthesis”. In: *Ecological Applications* 23.4, pp. 944–958. ISSN: 1051-0761. DOI: 10.1890/12-0854.1. URL: <http://www.esajournals.org/doi/abs/10.1890/12-0854.1>.
- Dietze, Michael C, David S. LeBauer, and Rob Kooper (Sept. 2013). “On improving the communication between models and data”. In: *Plant, Cell & Environment* 36.9, pp. 1575–1585. ISSN: 01407791. DOI: 10.1111/pce.12043. URL: <http://www.ncbi.nlm.nih.gov/pubmed/23181765>.
- LeBauer, David S, Michael C Dietze, and Benjamin M Bolker (2013). “Translating Probability Density Functions : From R to BUGS and Back Again”. In: *R Journal* 5.1, pp. 207–209. URL: <http://journal.r-project.org/archive/2013-1/lebauer-dietze-bolker.pdf>.
- LeBauer, David S (2010). “Litter degradation rate and β -glucosidase activity increase with fungal diversity”. In: *Canadian Journal of Forest Research* 40.6, pp. 1076–1085. ISSN: 0045-5067. DOI: 10.1139/X10-054. URL: <http://article.pubs.nrc-cnrc.gc.ca/ppv/RPViewDoc?issn=1208-6037%5C&volume=40%5C&issue=6%5C&startPage=1076%5C&ab=y>.
- Wang, Dan, David S LeBauer, and Michael C Dietze (2010). “A quantitative review comparing the yield of switchgrass in monocultures and mixtures in relation to climate and management factors”. In: *GCB Bioenergy* 2.1, pp. 16–25. ISSN: 17571693. DOI: 10.1111/j.1757-1707.2010.01035.x. URL: <http://doi.wiley.com/10.1111/j.1757-1707.2010.01035.x>.
- Allison, Steven D., David S. LeBauer, M. Rosario Ofrecio, Randy Reyes, Anh Minh Ta, and Tri M. Tran (Feb. 2009). “Low levels of nitrogen addition stimulate decomposition by boreal forest fungi”. In: *Soil Biology and Biochemistry* 41.2, pp. 293–302. ISSN: 00380717. DOI: 10.1016/j.soilbio.2008.10.032. URL: <http://linkinghub.elsevier.com/retrieve/pii/S0038071708003908>.

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- Dietze, Michael C., Shawn P. Serbin, Carl Davidson, Ankur R. Desai, Xiaohui Feng, Ryan Kelly, Rob Kooper, David S LeBauer, Joshua Mantooth, Kenton McHenry, and Dan Wang (2014). “A quantitative assessment of a terrestrial biosphere model’s data needs across North American biomes”. In: *Journal of Geophysical Research: Biogeosciences* 119.3, pp. 286–300. ISSN: 21698961. DOI: 10.1002/2013JG002392.
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Conference proceedings

- LeBauer, David, Max Burnette, Noah Fahlgren, Rob Kooper, Kenton McHenry, and Abby Stylianou (2021). “What Does TERRA-REF’s High Resolution, Multi Sensor Plant Sensing Public Domain Data Offer the Computer Vision Community?” In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 1409–1415.
- Pistorius, J., C. Martin, S. Sudarshan, and D. S. LeBauer (2020). “Exosphere - Bringing The Cloud Closer”. In: *2020 IEEE/ACM International Workshop on Interoperability of Supercomputing and Cloud Technologies (SuperCompCloud)*. IEEE, pp. 1–6. DOI: 10.1109/SuperCompCloud51944.2020.00006.
- Schnauffer, Christophe, Julian L Pistorius, and David S LeBauer (2020). “An open, scalable, and flexible framework for automated aerial measurement of field experiments”. In: *Proceedings of SPIE*. DOI: 10.1117/12.2560008.
- Burnette, Maxwell, Rob Kooper, JD Maloney, Gareth S Rohde, Jeffrey A Terstriep, Craig Willis, Noah Fahlgren, Todd Mockler, Maria Newcomb, Vasit Sagan, et al. (2018). “TERRA-REF Data Processing Infrastructure”. In: *Proceedings of the Practice and Experience on Advanced Research Computing*. ACM, p. 27.

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Data

- Lachowiec, Jennifer, Max Feldman, Filipe Inacio Matias, David LeBauer, and Alexander Gregory (n.d.). *Data and Code from a Survey of UAS for Phenotyping in Agriculture*. DOI: 10.5281/zenodo.10783692. URL: <https://github.com/Lachowiec-Lab/agDronesSurvey>.
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- David, Etienne, Mario Serouart, Daniel Smith, Simon Madec, Kaaviya Velumani, Shouyang Liu, Xu Wang, Francisco Pinto Espinosa, Shahameh Shafiee, Izzat S. A. Tahir, Hisashi Tsujimoto, Shuhei Nasuda, Bangyou Zheng, Norbert Kichgessner, Helge Aasen, Andreas Hund, Pouria Sadeghi-Tehran, Koichi Nagasawa, Goro Ishikawa, Sébastien Dandrifosse, Alexis Carlier, Benoit Mercatoris, Ken Kuroki, Haozhou Wang, Masanori Ishii, Minhajul A. Badhon, Curtis Pozniak, David Shaner LeBauer, Morten Lilimo, Jesse Poland, Scott C. Chapman, Benoit de Solan, Frédéric Baret, Ian Stavness, and Wei Guo (2021). “Global Wheat Head Dataset 2021: an update to improve the benchmarking wheat head localization with more diversity”. In: *CoRR* abs/2105.07660. arXiv: 2105.07660. URL: <https://arxiv.org/abs/2105.07660>.
- LeBauer, David, Maxwell A. Burnette, Jeffrey Demieville, Noah Fahlgren, Andrew N. French, Roman Garnett, Zhenbin Hu, Kimberly Huynh, Rob Kooper, Zongyang Li, Maitiniyazi Maimaitijiang, Jerome Mao, Todd C. Mockler, Geoffrey Morris, Maria Newcomb, Michael J Ottman, Philip Ozersky, Sidike Paheding, Duke Pauli, Robert Pless, Wei Qin, Kristina Riemer, Gareth Scott Rohde, William L. Rooney, Vasit Sagan, Nadia Shakoob, Abby Stylianou, Kelly Thorp, Richard Ward, Jeffrey W White, Craig Willis,

and Charles S Zender (2020). *TERRA-REF, An Open Reference Data Set From High Resolution Genomics, Phenomics, and Imaging Sensors*. *Dryad Digital Repository*. DOI: 10.5061/dryad.4b8gt99.

Anderson-Teixeira, Kristina, Maria Wang, Jennifer McGarvey, Valentie Herrmann, Alan Tepley, Ben Bond-Lamberty, and David LeBauer (2018). “ForC: A global database of forest carbon stocks and fluxes”. In: *Ecology*.

LeBauer, David S. and Kathleen K. Treseder (Aug. 2016). *Data and Code From "Nitrogen Limitation of Net Primary Productivity in Terrestrial Ecosystems is Globally Distributed"*. DOI: 10.6084/m9.figshare.c.3300164.v1. URL: <https://doi.org/10.6084/m9.figshare.c.3300164.v1>.

LeBauer, D, M Dietze, R Kooper, S Long, P Mulrooney, GS Rohde, and D Wang (2010). *Biofuel Ecophysiological Traits and Yields Database (BETYdb)*. URL: <https://betydb.org>.

Software

LeBauer, David, Max Burnette, Craig Willis, Zongyang Li, Robert Pless, Solmaz Hajmohammadi, Scott Rohde, and TERRA REF contributors (2017). *TERRA Reference Phenotyping Platform database and computing pipeline (terraref.org)*. code: github.com/terraref. URL: terraref.org.

Rohde, Scott, Patrick Mulroony, Rob Kooper, Carl Crott, Andrew Shirk, and David S LeBauer (2017). *Biofuel Ecophysiological Traits and Yields database (BETYdb, betydb.org)*. code: github.com/pecanproject/bety. Version 4.18. Database network and web interface to harmonize heterogeneous plant, ecosystem, and agronomic data. Supports meta-analysis, simulation modeling, and synchronization. Currently in use by six research teams. DOI: 10.5281/zenodo.832983. URL: betydb.org.

LeBauer, David, Michael Dietze, Rob Kooper, Alexey Shiklomanov, Betsy Cowdery, Istem Fer, Anthony Gardella, Ben Bond-Lamberty, Shawn P. Serbin, Ann Raiho, Anne Thomas, Chris Black, James Simkins, Ankur Desai, Josh Mantooth, Aman Kumar, Liam Burke, Afshin Pourmokhtarian, Christy Rollinson, Shubham Agarwal, Brady Hardiman, Martin De Kauwe, Tess McCabe, Tony Cohen, Tony Viskari, Yan Zhao, and Jing Xia (2017). *Predictive Ecosystem Analyzer (PEcAn, pecanproject.org)*. code: github.com/pecanproject/pecan. Version 1.5.1. Ecoinformatics toolbox for model-data synthesis, analysis, and prediction. Integrates observations with scientific understanding of crop and ecosystem functioning as an iterative, directed learning process. Increased efficiency of science over 100x by overcoming technical obstacles. Used by many crop and ecosystem modeling groups. DOI: 10.5281/zenodo.1003756. URL: pecanproject.org.

- Miguez, Fernando E, Deepak Jaiswal, David S LeBauer, Dan Wang, and Justin McGrath (2015). *BioCro Crop Productivity and Ecosystem Service Simulation Model*. code: github.com/ebimodeling/biocro. Version 0.93. Dynamical simulation model combining physics, chemistry and physiological processes to predict crop growth and water use. DOI: 10.5281/zenodo.15859. URL: github.com/ebimodeling/biocro.
- Anderson-Teixeira, Kristina, Evan DeLucia, Carl Crott, David S LeBauer, Scott Rohde, Nicholas Potter, and Jay Dorsey (2015). *Ecosystem Climate Regulation Services Calculator*. code: github.com/ebimodeling/ghgvc. Version 1.0. URL: ecosystemservicescalc.org.

Letters

- Jaiswal, Deepak, Amanda P De Souza, Søren Larsen, David S LeBauer, Fernando E Miguez, Gerd Sparovek, Germán Bollero, Marcos S Buckeridge, and Stephen P Long (2019). “Reply to: Brazilian ethanol expansion subject to limitations”. In: *Nature Climate Change* 9.3, p. 211.

Book chapter

- Isikhuemhen, O.S. and D.S. LeBauer (2004). “Growing *Pleurotus tuberregium*”. In: *Oyster Mushroom Cultivation*. Seoul, Korea: MushWorld, pp. 270–281.

Miscellaneous

- Riemer, Kristina, Kimberly. M. Huynh, Christophe Schnauffer, Julian Pistorius, Jorge. E. Barrios, Emily. J. Cain, Jessica Guo, Eric R. Scott, Renata Diaz, and David S. LeBauer (Mar. 2024). *CCT Data Science Group Procedures*. DOI: 10.5281/zenodo.10783692. URL: <https://github.com/cct-datascience/group-procedures>.
- LeBauer, David (2020). *Transforming ANOVA and Regression statistics for Meta-analysis*. DOI: 10.22541/au.158359749.96662550. URL: <https://doi.org/10.22541/au.158359749.96662550>.
- (2015). *Shiitake Cultivation on Logs*. URL: <https://www.overleaf.com/articles/shiitake-cultivation-on-logs/xphcccwyzhbb.pdf>.
- Tu, Andy and David LeBauer (2011). *Data Entry Workflow for the Biofuel Ecophysiological Traits and Yields Database*.

Invited talks

- 2023** Building a Data Science Support Team (YouTube). Case Studies in Data Science, UCSD Masters of Advanced Studies Data Science and Engineering Program, San Diego, CA, January.
- 2020** Software to Streamline Sharing of Agricultural Algorithms and Data. Challenges and Opportunities in Plant Science Data Management workshop, Plant and Animal Genome Conference, San Diego, CA, January.
- 2019** Agriculture in the Digital Age: Challenges and Opportunities. Department of Electrical and Computer Engineering Seminar, Gainesville, FL, March.
- 2019** Vocabularies, APIs, and Formats for Heterogeneous High Throughput Crop Phenotyping Data. Interoperability and Federation Across Bioinformatic Platforms and Resources, Plant and Animal Genome Conference, San Diego, CA, January.
- 2018** Open Software, Data, and Computing Frameworks to Advance High Throughput Phenomics and Agricultural Forecasting. Center for Sustainable Resource Science, RIKEN Yokohama Campus, Yokohama, Japan, February.
- 2018** Open Software, Data, and Computing Frameworks to Advance High Throughput Phenomics and Agricultural Forecasting. Data driven crop design technology. Institute of Mathematics for Industry, Kyushu University, Fukuoka, Japan, February.
- 2018** Software and Data as Scaffolds for Integrative Science. Department of Agricultural and Biological Engineering, University of Arizona, Tucson, AZ, February.
- 2017** Panelist. Agricultural Data Integration: From Genomics to Unmanned Systems. Institute of Mathematics and its Applications, University of Minnesota, Minneapolis, MN, October.
- 2017** TERRA REF Data and Computing Infrastructure. DARPA Data Driven Discovery of Models (D3M) Fall Hackathon and Evaluation Dry-Run, Arlington, VA, October.
- 2017** TERRA REF Data and Computing Infrastructure.. ARPA-E Joint TERRA and ROOTS Annual Meeting, Saint Louis, MO, September.
- 2017** Software and Data as Scaffolds for Integrative Science.. College of Agriculture, Purdue University, West Lafayette, IN, August.

- 2017** A Meteorological Data Format That Makes Research Easier and Will Work for Imaging Sensors.. Phenome 2017, Tucson, AZ, February.
- 2016** Building Open Access Data, Computing, and Software Infrastructure for High Throughput Phenomics. Plant Imaging Consortium Annual Meeting, University of Arkansas, Fayetteville, AK, July.
- 2016** An open access crop observatory and computing platform for plant breeding. ROGER Users Meeting, National Center for Supercomputing Applications, Urbana, IL, May.
- 2016** Uncertainty Analysis in the Predictive Ecosystem Analyzer (PEcAn). PEcAn 2 Terrestrial Model Informatics Workshop, Boston University, Boston, MA, May.
- 2016** Opportunities to Improve Cropping Systems with Data. International Food Security at Illinois Symposium: Using Big Data to Improve International Food Security, University of Illinois, Urbana, IL, April.
- 2016** Demonstration of The TERRA Phenotyping Reference Platform. 5th National Data Service Consortium Workshop, Chapel Hill, NC, April.
- 2015** An open-access platform for quantification of plant traits from big sensor data. Informatics for Reproducibility in Earth and Environmental Science Research, American Geophysical Union, San Francisco, CA, December.
- 2015** Software for Ecological Inference and Prediction: What We Have and What We Need. Developing, Maintaining, and Employing Large Computational Frameworks in the Ecological Sciences. Statistical and Applied Mathematical Sciences Institute, Durham, NC, April.
- 2014** Modeling the Productivity and Ecosystem Services of Sugarcane. Global Sustainable Bioenergy and FAPESP annual meeting, Piracicaba, Brazil, August.
- 2014** PEcAn: A probabilistic modeling workflow for deterministic models. FACE-IT Workshop, Computation Institute, University of Chicago, Chicago, IL.
- 2013** Modeling Bioenergy Feedstock Productivity and Ecosystem Services. Global Bioenergy Crop Modeling Workshop, Oak Ridge National Laboratory, Oak Ridge, TN.
- 2013** Reducing uncertainty through data-driven model development. Uncertainty Analysis: A Critical Step in Ecological Synthesis, Organized Oral Session, Ecological Society of America Meetings, Minneapolis, MN.

- 2010** PEcAn, a workflow management tool for real-time data assimilation and forecasting. Combining Experiments, Process Studies, and Models to Forecast the Future of Ecosystems, Communities, and Populations. Organized Oral Session, Ecological Society of America Meetings, Pittsburgh, PA.

Grants

- 2024–2026** California Cropland Carbon Monitoring and Modeling Framework. California Air Resources Board. Role: co-PI, Modeling Workflow Lead, Project Manager. (\$1.3m)
- 2022–2023** Broadening Diversity in the North American Plant Phenotyping Network. Agricultural Genome to Phenome Initiative (AG2PI). Role: PI. (\$15k)
- 2021–2022** Empowering High-Throughput Phenotyping using Unoccupied Aerial Vehicles (UAVs). Agricultural Genome to Phenome Initiative (AG2PI). Role: co-PI. (\$20k)
- 2021–2022** Evaluating High Bandwidth Satellite Communications for Data Intensive Agriculture. Yuma Center for Desert Agriculture Small Grants Program. Role: PI. (\$20k)
- 2019–2022** The Agricultural Research Data Network (ARDN). USDA-NIFA. Role: Co-PI. (\$75k of \$500k)
- 2018–2021** SENTINEL: SENSing Threats In Natural Environments using Ligand-receptor modules. DARPA. Role: Co-PI. (\$731k of ~\$10m)
- 2016–2019** Global Sustainable Bioenergy Initiative: Geospatial and environmental analysis of pastureland intensification for bioenergy. FAPESP (Fundacao de Amparo a Pesquisa do Estado de Sao Paulo). Role: Co-PI. (\$200k)
- 2016–2019** TERRA REF Computing and Storage Allocation. National Center for Supercomputing Applications: Blue Waters project and CyberGIS Center. Role: PI, Data and Computing Pipeline Lead. (\$1m+ in computing resources)
- 2015–2019** A Reference Phenotyping System for Energy Sorghum. U.S. Department of Energy: Advanced Research Projects Agency - Energy (ARPA-E). Role: Co-PI. (\$1.7m of \$8.4m)
- 2015–2017** TERRA-MEPP (Mobile Energy-crop Phenotyping Platform). U.S. Department of Energy: Advanced Research Projects Agency - Energy (ARPA-E). Role: Co-PI. (\$132k of \$3.4m)

- 2014–2014** Advancing Software for Ecological Forecasting (Workshop). Microsoft Research Connections, Carl R Woese Institute for Genomic Biology, DOE RCN Forecast. Role: Chair and Co-PI. (\$30k)
- 2012–2015** Feedstock and Ecosystem Service Modeling Program. Energy Biosciences Institute. Role: Lead author and program manager. (\$2.5m)
- 2011–2014** Model-data synthesis and forecasting across the upper Midwest; Partitioning uncertainty and environmental heterogeneity in ecosystem carbon.. NSF Advances in Bioinformatics Infrastructure. Role: Co-author. (\$770k)

Awards

- 2023–2023** Equity, Inclusion, and Diversity Award. North American Plant Phenotyping Network
- 2021–2021** Best Paper. 7th workshop on Computer Vision in Plant Phenotyping and Agriculture. ICCV 2021
- 2017–2017** Outstanding Students Pushing Innovation (SPIN) Mentor. National Center for Supercomputing Applications
- 2014–2015** NCSA Fellowship. National Center for Supercomputing Applications (\$25k)
- 2006–2006** Mildred E. Mathias Graduate Research Grant, UC Natural Reserve System. Decomposition responses to nitrogen in a California grassland (\$1k)
- 2005–2007** Graduate Fellowship. Kearney Soil Science Foundation (\$34k)
- 1998–1998** Honors in Ecology and Graduation with Distinction. Duke University Department of Biology
- 1998–1998** Benenson Award in the Arts. Duke University (\$1.5k)

Teaching

- 2021** Open Science Synthesis for the Delta Science Program, National Center for Ecological Analysis and Synthesis. Davis, CA
- 2018–present** CCT Data Science Workshops. [\url{datascience.cals.arizona.edu/workshops}](https://datascience.cals.arizona.edu/workshops)
- 2017** Data Science in Modern Agriculture (Graduate Seminar), University of Illinois Department of Agricultural and Biological Engineering, Urbana, IL
- 2017–2018** Instructor, Computational Mathematics Bootcamp, Program for Interdisciplinary and Industrial Internships at Illinois, Urbana, IL. pi4-uiuc.github.io

io/2017-bootcamp

- 2017** Instructor, Advanced Phenomics Workshop, Purdue University Departments of Agronomy and Agricultural and Biological Engineering, West Lafayette, IN.
- 2015–2018** Instructor, Urbana Math Circle, Urbana, IL. urbanamathcircle.blogspot.com
- 2015–present** Software Carpentry Instructor
- 2015** Instructor, Shiitake Cultivation on Logs, The Land Connection, Urbana, IL
- 2013–2014** Instructor, PEcAn Software Training, Ecological Society of America Meeting
- 2009** Instructor, Global Change Biology, California Summer School for Science and Math
- 2008** TA, Experimental Biology Lab, UC Irvine
- 2008** TA, Atmospheric and Environmental Sciences, California Summer School for Science and Math
- 2008** TA, GIS for Environmental Science, UC Irvine
- 2006** TA, GIS for Environmental Science, UC Irvine
- 2007** TA, Global Change Biology, UC Irvine
- 2003** Instructor, Shiitake Cultivation Workshops, Hosted by NC Extension at multiple locations
- 2003** TA, Mushroom Cultivation, UC Davis
- 1997** TA, Archaeology and Geology Southwest field trip, Duke Talent Identification Program

Service

Academic

- 2021** Co-organizer, University of Arizona Drone Day [\url{drones.arizona.edu/ua-drone-day-2022}](http://drones.arizona.edu/ua-drone-day-2022).
- 2021–2022** Organizing Committee, North American Plant Phenotyping Network Conference.
- 2020–2023** Member, University of Arizona Data Science Resources and Training Steering Committee.

2017–present Google Summer of Code Mentor, PEcAn Project.

2015–2018 Undergraduate Student Mentor, Students Pushing Innovation Program, National Center For Supercomputing Applications.

2015–2016 Reviewer, EarthCube Architecture Conceptual Design.

2014–2016 Graduate Student Summer Intern Mentor, Program for Interdisciplinary and Industrial Internships at Illinois.

2012–2016 Campus Representative, National Ecological Observatory Network (NEON).

2009–2011 Expert for Climate Science Experts Referral Service, American Geophysical Union.

2006–2008 Judge, California State Science Fair.

2006–2007 Graduate Student Representative, UC Irvine Department of Earth System Science.

2005–2006 Graduate Student Seminar Organizer, UC Irvine Department of Earth System Science.

2003 Project Mentor, FARMS Leadership, Inc. and Sacramento Public Schools.

1998–1999 Guest Teacher, K-12, Sigma Xi, Scientists in the Classroom.

Workshops

February 2019 Organizer. PEcAn+CLM Coupling and Beyond Workshop and Hackathon, Santa Rita Experimental Range, Green Valley, Arizona.

February 2018 Co-organizer. Big Data Driven Agriculture: Advances, Challenges, and Opportunities, USDA NIFA-FACT, Alexandria, VA.

November 2017 Invited participant. The Subterranean Macroscopic: Sensor Networks for Understanding, Modeling, and Managing Soil Processes, NSF, Chicago, IL.

February 2017 Co-organizer, lead instructor. Using TERRA REF phenotyping data, Phenome 2017, Tucson, AZ.

February 2017 Co-organizer. Phenome 2017 post-conference hackathon, University of Arizona, Tucson, AZ.

January 2017 Invited Participant. Moving Field Phenomics From Theory to Practice, US-UK Phenomics Workshop, Maricopa, AZ.

October 2016 Invited Participant. Plant Science Cyberinfrastructure Initiative, Plant Science Research Network, Washington, DC.

August 2016 Session Co-Organizer. Hacking Ecology 2.0, a showcase of open tools for data-driven ecology, Ecological Society of America Annual Meetings, Fort Lauderdale, FL.

November 2014 Workshop Co-Organizer. Soil Carbon Cycle Super-Modeling, Biosphere 2, Oracle, AZ.

August 2014 Symposium Co-Organizer. Challenges and Advances in Statistical Software For Ecology., Ecological Society of America Annual Meetings, Sacramento, CA.

March 2014 Workshop Chair and Co-Organizer. Advancing Software for Ecological Forecasting, Urbana, IL.

Grant Reviews

United States Department of Agriculture
US National Science Foundation
Foundation for Food and Agricultural Research

Journal Reviewer

Applied Soil Ecology
Ecology
Ecology Letters
Ecological Applications
Frontiers in Plant Science (Review Editor)
Global Change Biology
Global Change Biology - Bioenergy
Geophysical Research Letters
Global Ecology and Biogeography
Journal of Geophysical Research
Journal of Geophysical Research - Biogeosciences
Journal of Open Source Software
Journal of Open Source Education

Nature Climate Change
 Nature Scientific Reports
 New Phytologist
 Open Data Journal for Agricultural Research
 Plants, People, Planet
 Proceedings of the National Academy of Sciences of the United States of America
 Plant Soil

Consultancy

2015–2018 Agrible, Inc., Champaign, IL
2015–2016 Smithsonian Institution, CTFS – ForestGEO Ecosystems and Climate Initiative, Falls Church, Va
2011–2014 Global Change Solutions, LLC, Urbana, IL
1999–2000 Wildflower Organics, Dawsonville, GA
2000 Crabtree Valley Farms, Chatanooga, TN
1999 Sustenance Farm, Bear Creek, NC

Students

2009–2009 Sen Lu, *Meta-analysis of nitrogen fertilization effects on productivity over time.*
 Independent Study
2008–2008 Elizabeth Chandler, *Minimally invasive monitoring of grassland carbon inputs in a long term nitrogen addition experiment.*
 Kearney Undergraduate Fellowship (\$3,000); faculty advisor Sue Trumbore
2007–2007 Elizabeth Chandler, *The Effects of Nitrogen Addition at Different Stages of Decomposition (UCI Honors Thesis)*
 Kearney Undergraduate Fellowship (\$3,000); faculty advisor Kathleen Treseder